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Machine Safety Specialist (CE)

Module 1: Principles of the Work Equipment Directive

TAB5: Work Equipment Directive
Tab5-1/1 20221202

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Tab5-1/1-Sh.2

Subdivision Directives

Product directives art. 114 (CE)	Social directives art. 153 (Arbo)
• Machinery Directive • EMC Directive • Low Voltage Directive • Pressure Equipment Directive • ATEX Directive • Gas Appliances Directive • Personal Protective Equipment Directive	• Work Equipment Directive* • Manual handling of loads directive • Directive on safety and/or health signs at work • Temporary and mobile construction sites directive • VDU Work Directive*
Included in the Commodities Act in the Netherlands.	Included in the Working Conditions legislation in the Netherlands

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Tab5-1/1-Sh.3

Working Conditions Act makes sense?

Tab5-1/1-Sh.4

Work Equipment Directive

2009/104/EG

- Previously: 2001/45/EG (19-07-2004), 95/63/EG (5-12-2002), 89/655/EEG, mandatory by 1-1-1997
- Included in the Working Conditions Act (Working Conditions Decree Chapter 7)

Tab5-1/1-Sh.5

Definition of work equipment

(article 2)

- All machinery, equipment, tools and installations used at the workplace

Tab5-1/1-Sh.6

Basic idea

The safety level of work equipment must be guaranteed during all phases of its lifespan:

- Transport
- Installation and adjustment
- Use, Setup
- Operation, cleaning
- Programming, maintenance
- Decommissioning, demolition and disposal

Tab5-1/1-Sh.7

Structure of the Work Equipment Directive

Considerations

Articles

Attachments

1. Minimum requirements
2. Additional requirements

Tab5-1/1-Sh.8

Step-by-step plan for the Work Equipment Directive

Risk inventory and evaluation

1. Inventory of work equipment
2. Assessment of minimum requirements
3. Risk assessment
   - Dangers
   - Risk analysis
   - Risk assessment
4. Risk reduction (testing against standards)
5. Drawing up and implementing an action plan

Periodical

6. Inspections and tests
7. Drawing up a maintenance plan
8. Organizational obligations
   - Procedures, instructions
   - Information, training
   - Supervision

General minimum requirements

Attachment 1

- Service
- Start/stop
- Strength/stability
- Moving parts
- Lighting
- Temperature
- Alarm
- Maintenance, installation, (dis)assembly
- Warnings
- Fire/explosion
- Electricity

Question 5-1

What are the differences and similarities with the subjects on which the Machinery Directive sets requirements in Annex I??

- Service
- Start/stop
- Strength/stability
- Moving parts
- Lighting
- Temperature
- Alarm
- Maintenance, installation, (dis)assembly
- Warnings
- Fire/explosion
- Electricity

Tab5-1/1-Sh.11

Additional minimum requirements for mobile work equipment (Working Conditions Decree Div. 4 par. 2)

- Ride along
- Energy transmission
- Stability
- Collision
- Running
- Braking/stopping
- Sight
- Lighting
- Education
- People on foot in work zone
- Combustion engine

Tab5-1/1-Sh.12

Additional minimum requirements for work equipment used for lifting and hoisting loads or persons (Working Conditions Decree Div. 4 par. 2a)

- Strength and stability
- Nominal load indication
- Hoisting and lifting accessories
- Falling/colliding loads
- Hoisting/lifting persons
- Communication
- Evacuation
- Operation and supervision
- Performing work
- Coordination/procedures
- Power supply failure
- Weather

Tab5-1/1-Sh.13

Additional minimum requirements for temporary work at height (Working Conditions Decree Div. 4 par. 2b)

- Suitable resources
- (Separate) ladder can be used to a limited extent
- Positioning ladder
- Fall protection devices
- Weather
- Scaffolding
- Working line/safety line (independent)

Tab5-1/1-Sh.14

Framework directive

89/391/EEG

Purpose :

- To improve the safety and health of workers. The employer is obliged to ensure the safety and health of the employees.

The employer is always responsible, even if third parties are called upon.

Included in the Working Conditions Act, various decrees

Tab5-1/1-Sh.15

Framework directive:
Employer's obligations

- Performing a risk inventory and evaluation
- Drawing up an action plan
- Adaptation of work (ergonomics)
- Taking into account technical developments
- Replacement of hazardous situations with less dangerous situations
- Planning of prevention measures (Arbozorg))

Tab5-1/1-Sh.16

Framework directive:
Employer's obligations

- Organization, policy, consultation structure, procedures, instructions
- Company emergency plan, emergency response
- Appointing a prevention officer
- Information and training
- Participation
- Monitoring risks in the workplace
- Opportunity for regular medical examinations

Tab5-1/1-Sh.17

Framework directive:

Obligations of the employee

- Use work equipment and hazardous substances correctly;
- To use the personal protective equipment made available to him correctly;
- Not to modify or remove the safety devices installed on work equipment or otherwise without necessity and to use them properly;
- To cooperate in the instruction organized for him;
- Report the safety or health hazards he notices to the employer or manager;
- Assist the employer and the employees, other persons and services, if necessary, in the performance of their obligations and duties under this Act.

Tab5-1/1-Sh.18

RI&E

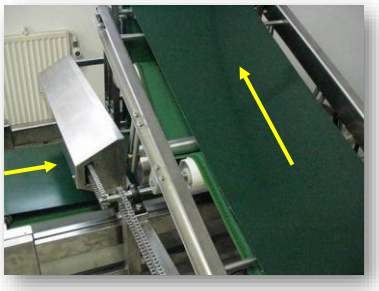
- Legal obligation (Working Conditions Act; Article 5)
- Pay attention to the health and safety of employees
- Attention to special groups of people
- Industrial accidents and absenteeism
- Depth at:
  - Hazardous substances
  - ionising radiation
  - Sound
  - Physical strain
  - VDU work
  - Personal protective equipment
  - accidents, occupational diseases, absenteeism
- Plan of action revised annually
- Risk assessment of work equipment is an integral part

Tab5-1/1-Sh.19

Exercise 5-2

- Test the following situation against the literal Minimum Requirements text. Does the work equipment comply with:

Work Equipment Directive, par. 2.8 (Working Conditions Decree 7.7):
"Where moving parts of work equipment are liable to come into contact with mechanical accidents, they must be fitted with screens or devices designed to prevent access to danger zones or to stop the movement of dangerous parts before they reach them.."



- The area around the discharge belts may be disregarded.
- Both conveyor belts are accessible.

Tab5-1/1-Sh.20

Exercise 5-3

- Test the following situation against the literal Minimum Requirements text.
- Does the work equipment comply with:

Work Equipment Directive, par. 2.4 (Working Conditions Decree 7.16):
"If this is necessary in view of the hazards of the work equipment and the normal switch-off time, work equipment must be equipped with an emergency stop device."

- The green button on the control panel of the press brake works according to the control requirements of an emergency stop device.

Tab5-1/1-Sh.21

Dutch working conditions legislation

- Working Conditions Act (part of the Civil Code)
- Working Conditions Decree
- Working Conditions Regulations

- Health and Safety Catalogue
- A(rbo)I(nformation)sheets
- European standards

Tab5-1/1-Sh.22

Working Conditions Decree

Articles at chapter level from the Working Conditions Decree(10-2016):

1. Definitions and scope
2. Occupational health and safety care and organisation of work
3. Layout of workplaces
4. Dangerous substances and biological agents
5. Physical strain
6. Physical factors
7. Work equipment and specific work
8. Personal protective equipment and safety and health signage.
9. Obligations, criminal offences, fineable offences, administrative provisions and transitional and final provisions

Health and safety catalogue

- Per section
- Purpose regulations (determine how to comply with Legislation)
- Approval by the Inspectorate of Social Affairs and Employment (marginal review)
- Not mandatory, but desirable
- Compliance lies with companies
- Control at Inspection

Check out: www.arboportaal.nl

Inspections in the use phase

Working Conditions Decree article 7.4;
Inspections
Applies:

1. After installation and before commissioning (e.g. after re-assembly)
2. If work equipment is subject to influences that lead to deterioration that can lead to the occurrence of dangerous situations
3. If exceptional events have occurred which may have harmful effects on the safety of the work equipment

Examples:

- Scaffolding
- Hoisting, lifting equipment and tools
- Mobile work equipment
- Pressure equipment
- Electrical installations
- Warehouse equipment

Tab5-1/1-Sh.25

Maintenance

Working Conditions Decree article 7.5.
Assembly, disassembly, maintenance, repair and cleaning of work equipment.

1. Ensure that the work equipment is kept in such a condition throughout its useful life by adequate maintenance that it does not endanger the safety and health of workers.

2. Work on work equipment is only carried out when the work equipment has been switched off and depressurised or de-energised.

3. A maintenance book associated with work equipment is kept up to date.

Tab5-1/1-Sh.26

Information and training

Working Conditions Decree article 7.11a

1. Instructions for use accompanying work equipment shall be brought to the attention of the workers concerned in an intelligible form.

2. If the use or presence of work equipment in the immediate vicinity of the work may present dangers to workers, they shall be made aware of this, even if the workers do not make direct use of such equipment.

Purchasing a machine

(link between product directives and work equipment directive)

Working Conditions Decree article 7.2

Article 7.2 prohibits the employer from making work equipment available without CE marking and the accompanying EC declaration of conformity when they should have been available on the basis of an applicable Commodities Act Decree.

Work equipment with CE marking and an EC Declaration of Conformity is presumed to comply with the requirements relating to::

- Virtue
- Safety features for moving parts
- Alarms
- Control systems
- Start-up, shutdown and emergency stop devices

Include this work equipment in the RI&E and check the condition and documentation upon entering the machine.

Exercise 5-4

Answer the questions below:

1. What is meant by work equipment?
2. Name a number of topics (regulations) that are covered in the Work Equipment Directive.
3. Where in the Dutch law can the Work Equipment Directive be found?
4. The Work Equipment Directive is a specific Directive for Work Equipment. What is the overarching Directive ?

Tab5-1/1-Sh.29

Risk assessment of work equipment

Can be done via:

- Risk Graph Method (ISO/TR 14121-2)
- Fine & Kinney method
- HAZOP
- TRA (Human Actions)
- etc.

Tab5-1/1-Sh.30

Risk reduction according to Working Conditions Act H2, Article3-1.b

The "Occupational Hygiene Strategy""

1. *Source measures*
   An employer must first remove the cause of the problem. For example: replacing a harmful substance with a safer alternative.

2. *Collective (technical) measures*
   If source-based measures do not offer any options, the employer must take collective measures to reduce risks, for example: installing shielding or an extraction system.

3. *Individual (organisational) measures*
   If collective measures are not possible, or do not provide an adequate solution, the employer must take organisational measures to protect individual workers. For example: by exposing employees to the risk for a shorter period of time through more task rotation.

4. *Personal protective equipment*
   If the top three measures have no effect, the employer must provide the employee with personal protective equipment (free of charge). For example: hearing protection and/or safety shoes.

https://wetten.overheid.nl/BWBR0010346/2022-05-20#Hoofdstuk2

Tab5-1/1-Sh.31

Step 1: Source measures

Avoiding/removing danger (source) by, for example,:
- Replaced fabrics
- Removing the sound source

But also:

• Safety distances	EN 13854
• Ergonomics	EN 614, 547, 1005
• Design	EN-ISO 4413, 4414, 12100
• Operation	EN 894
• Restart, energy	EN-ISO 14119

Tab5-1/1-Sh.32

Step 2: Collective (technical) measures

If the source (danger) cannot be removed:

• Guards	EN-ISO 14120
• Two-hand control	EN-ISO 13851
• Light curtain	EN 61496-1
• Guard locking	EN-ISO 14119
• Safety functions	EC-ISO 13849-1 / EN-IEC 62061
• Emergency stop devices	EN-ISO 13850
• Provisions for rescue	
• of obstructed persons	
• Transport facilities	
• Means of access	EN-ISO 14122-1 t/m 4

Tab5-1/1-Sh.33

Step 3: Individual (organizational) measures

If collective action is not possible:

- Task rotation

- Safety and health signage
  - Alarmsignals — EN 61310-1, EN 981
  - Icons — EN 61310-2 & EN ISO 7010
  - Text warnings — EN 61310-2

Tab5-1/1-Sh.34

Step 4: Personal protective equipment

If measures in the first 3 steps have no effect:

- Personal protective equipment such as
  - Hearing protection
  - Respiratory protection
  - Safety (hand) gloves

Tab5-1/1-Sh.35

Reasonability principle

"The reasonability principle means that as an employer you are never obliged to take measures that cannot reasonably be expected of you.

- Emphatically a hierarchical order
- Only allowed to lower a level if there are good reasons for doing so (technical, executive, economic)
- Disproportionately expensive alternatives are not very realistic and therefore do not need to be investigated

- Exceptions: carcinogens, biological agents and mutagens and reprotoxic substances. Only lower if a higher level is not technically possible.

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